SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.4: Congruence — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.4: Congruence
Driving Question	DRIVING QUESTION: "How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy — and how can we use that same idea to prove that two shapes are truly identical?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Boat and Its Reflection: Introducing Reflection as a Transformation**	Students observed the anchoring phenomenon: photographs and diagrams of boats on Lake Victoria alongside their water reflections. They noted that the reflected image appears flipped but identical in size and shape to the original boat. Students also explored physical mirrors and reflected cut-out shapes to gather initial evidence about what reflection does to a figure.	Students learned that reflection is a type of geometric transformation in which every point of a figure is mapped to a corresponding image point on the opposite side of a fixed line (the mirror line / line of reflection), at exactly the same perpendicular distance from that line. The mirror line acts as a perpendicular bisector of the segment joining each object point to its image point. Reflected images preserve size and shape — a key observation that seeds the concept of congruence in later lessons.	The teacher should emphasise that 'flipping' is the informal term students bring in; the formal mathematical term is reflection. Use the Lake Victoria boat context consistently: the water surface acts as the mirror line. Draw attention to three critical properties that students will build on throughout the unit — (1) distance from mirror is preserved, (2) the connecting segment is perpendicular to the mirror line, and (3) the image is the same size as the object. Pedagogical note: cold-call at least three students to state in their own words what the mirror line does to a point, before formalising the definition. Allow 30 seconds of wait time after each prompt. Post the three properties on the Driving Question Board (DQB) as 'things we now know' to anchor future lessons.
Lesson 2 **Flipping Across the Axes: Reflection on the Cartesian Plane (x- and y-axes)**	Students plotted coordinates of a boat-shaped polygon on the Cartesian plane and systematically reflected its vertices across the x-axis and then across the y-axis. They recorded the original coordinates and image coordinates in a table, looking for numerical patterns between each object point and its	Students learned that reflecting a point across the x-axis maps $(x, y) \rightarrow (x, -y)$: the x-coordinate stays the same and the y-coordinate changes sign. Reflecting a point across the y-axis maps $(x, y) \rightarrow (-x, y)$: the y-coordinate stays the same and the x-coordinate changes sign. In both cases, the	The teacher should connect the coordinate rules back to the Lake Victoria phenomenon: reflecting the boat across the x-axis is equivalent to the water surface acting as a horizontal mirror. Stress that the sign-change rule is not arbitrary — it is a direct consequence of the perpendicular-

	image point.	image point is exactly the same perpendicular distance from the axis as the object point, confirming the distance-preservation property from Lesson 1.	bisector property of the mirror line. Pedagogical note: have students verify the rules by measuring distances from the axis to both the object and image points using a ruler on their printed grids. Cold-call four students — two for the x-axis rule and two for the y-axis rule — asking them to articulate WHY the sign changes, not just WHAT changes. Common misconception to address: students often swap both signs; emphasise which coordinate corresponds to which axis.
Lesson 3 **Reflecting Across Lines of the Form x = k and y = k**	Students observed what happens when the mirror line is not one of the coordinate axes but a vertical line such as $x = 3$ or a horizontal line such as $y = -2$. They plotted object points, drew the specified mirror line, and located image points by counting equal perpendicular distances on either side of the line.	Students learned that reflecting a point (x, y) across the vertical line $x = k$ maps it to $(2k - x, y)$: only the x-coordinate changes, and the image is as far to the right of the mirror as the object is to the left (or vice versa). Reflecting across the horizontal line $y = k$ maps $(x, y) \rightarrow (x, 2k - y)$: only the y-coordinate changes by the same logic. These rules are generalisations of the axis-reflection rules from Lesson 2 (where $k = 0$).	The teacher should help students see that $x = 0$ and $y = 0$ (the axes) are simply special cases of $x = k$ and $y = k$ with $k = 0$, unifying Lessons 2 and 3 into one coherent framework. Use the Lake Victoria context: if the boat is not centred on the shoreline ($y = 0$) but moored near a jetty at $y = 2$, the reflection rule shifts accordingly. Pedagogical note: ask students to derive the formula $2k - x$ rather than memorising it, by working from the definition that the mirror line bisects the object-image segment. Allow 45 seconds of think time before cold-calling two students to present their derivation. Address the common error of subtracting x from k rather than computing $2k - x$.
Lesson 4 **Diagonal Mirrors: Reflection Along y = x and y = −x**	Students reflected points and polygons across the diagonal lines $y = x$ and $y = -x$ on the Cartesian plane. They recorded coordinate pairs before and after reflection and looked for the algebraic pattern produced by each diagonal mirror line.	Students learned that reflecting a point across $y = x$ follows the rule $(x, y) \rightarrow (y, x)$: the coordinates are simply swapped. Reflecting a point across $y = -x$ follows the rule $(x, y) \rightarrow (-y, -x)$: the coordinates are swapped and both signs are changed. Both diagonal reflections preserve perpendicular distance from the mirror line and maintain the same size and shape of the reflected figure.	The teacher should encourage students to verify the $y = x$ rule using a physical 45° mirror or by folding a coordinate-grid sheet along the $y = x$ line and observing how plotted points land. Connect to the Lake Victoria phenomenon: a diagonal shoreline or a tilted mirror surface would produce these diagonal reflections. Pedagogical note: the swap rule for $y = x$ is highly intuitive once students fold the grid — use this tactile experience before algebraic formalisation. Cold-call three students to explain WHY swapping coordinates works for $y = x$ using the perpendicular-bisector definition. For $y = -x$, explicitly contrast it with $y = x$ to prevent students from applying only the swap rule and forgetting the sign

			change.
Lesson 5 **Reverse Engineering the Mirror: Finding the Equation of the Mirror Line**	Students were given pairs of congruent figures (object and image) on the Cartesian plane — without being told the mirror line — and tasked with locating the line of reflection by analysing the relationship between corresponding points. They drew segments joining object-image pairs and looked for a common bisector.	Students learned that the line of reflection can be determined from an object-image pair alone, without being told where the mirror is. The mirror line is the perpendicular bisector of the segment joining any object point to its corresponding image point. To find it: (1) calculate the midpoint of the segment — this lies on the mirror line; (2) calculate the gradient of the segment — the mirror line has the negative reciprocal gradient (since it is perpendicular); (3) use the midpoint and perpendicular gradient to write the equation of the mirror line.	The teacher should frame this as 'detective work': given only the boat and its reflection on Lake Victoria, can we locate exactly where the water surface (mirror line) lies? This reverse-engineering framing strongly motivates the skill. Pedagogical note: structure the lesson around a three-step algorithm (midpoint → gradient → equation) and have students write it in their own words before applying it. Cold-call four students at different steps of the algorithm during worked examples. Watch for the common error of using the gradient of the object-image segment rather than its negative reciprocal; address this explicitly using the definition of perpendicularity. Reinforce that this skill is the inverse of what students have been doing in Lessons 2–4.
Lesson 6 **Same Shape, Same Size: Introducing Congruence**	Students compared the measurements (side lengths, angles, perimeters, areas) of reflected figures and their original objects across multiple examples from previous lessons. They recorded findings in a structured comparison table, noting which properties were preserved and which (if any) were not.	Students learned that reflection always produces an image that is congruent to the original figure, because every point and its image are the same perpendicular distance from the mirror line — so all lengths, angles, perimeters, and areas are exactly preserved. Two figures are congruent (≅) if one can be mapped onto the other by a reflection (or other isometry) such that all corresponding sides are equal and all corresponding angles are equal. Congruence is the formal mathematical expression of the 'perfect, same-sized copy' seen in the Lake Victoria boat reflection.	The teacher should explicitly name this lesson as the conceptual heart of the unit: everything studied in Lessons 1–5 was building toward this formal definition of congruence. Use the language of correspondence: when stating congruence, the order of vertices matters (e.g., $\triangle ABC \cong \triangle DEF$ means $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$). Pedagogical note: have students return to their Lesson 1 DQB entries and annotate them with the word 'congruent' wherever they previously wrote 'same size and shape.' This metacognitive move reinforces conceptual progression. Cold-call three students to articulate the difference between figures that look congruent and figures that are proven congruent using the reflection mapping. Emphasise that congruence must be justified, not just observed.
Lesson 7 **Proving Congruence: Conditions and Formal Justification**	Students examined pairs of triangles and quadrilaterals, some congruent and some not, and attempted to determine congruence by checking various combinations of sides and angles. They	Students learned the standard congruence conditions for triangles: SSS (three sides), SAS (two sides and the included angle), ASA (two angles and the included side), and AAS (two angles and a non-included	The teacher should connect congruence conditions back to the reflection framework: when two triangles are reflections of each other, ALL sides and ALL angles correspond — so any of the sufficient

	tested whether knowing certain combinations (e.g., three sides, two sides and the included angle) was always sufficient to confirm congruence.	side). These conditions provide the minimum information needed to guarantee congruence without checking every measurement. RHS (right angle, hypotenuse, side) was introduced as an additional condition for right-angled triangles. Students also learned that SSA and AAA are generally NOT sufficient conditions for congruence.	conditions (SSS, SAS, ASA, AAS) will be satisfied. The conditions simply tell us the minimum evidence required to be certain. Pedagogical note: use physical cut-out triangles (labelled with Kenyan place names on their vertices, e.g., Nairobi, Kisumu, Mombasa) that students can manipulate and overlay to test each condition. Cold-call five students during the exploration — one per condition — asking them to state which measurements they checked and why those were sufficient. Explicitly demonstrate why AAA fails using two similar but non-congruent triangles, and why SSA is ambiguous using the 'ambiguous case' construction.
Lesson 8 **Mathematics in the Real World: Applications of Reflection and Congruence**	Students examined real-world contexts in which reflection and congruence appear in Kenyan settings: tiling patterns in Nairobi buildings, Kanga fabric designs with reflective symmetry, symmetric bridge structures in the Rift Valley, and satellite images of symmetric agricultural plots near Kericho. They identified mirror lines, corresponding congruent parts, and applied the congruence conditions from Lesson 7.	Students learned that: (1) Reflection maps every point in a figure to a corresponding point on the opposite side of the line of reflection, at exactly the same perpendicular distance — producing an image congruent to the original. (2) The equation of the mirror line can be determined from any object-image pair using midpoints and perpendicular gradients. (3) Congruence is formally verified using SSS, SAS, ASA, AAS, or RHS conditions — not just visual inspection. (4) These mathematical ideas underpin real-world design, engineering, and art in Kenyan contexts and beyond, answering the unit's driving question completely.	The teacher should use this lesson as a formal unit synthesis, explicitly returning to the driving question displayed since Lesson 1: 'How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy?' Have students construct a written response to the driving question using mathematical vocabulary from all eight lessons. Pedagogical note: conduct a gallery walk in which student groups display their model revisions from across the unit, annotated with reflection rules, congruence conditions, and real-world connections. Cold-call three students to connect one specific real-world application to one specific lesson concept. Close by updating the DQB one final time — moving all remaining questions to 'answered' and celebrating the conceptual journey from informal observation (Lesson 1) to formal proof (Lessons 6–8).